

Haar Measure on Topological Groups

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Figure 1: Claude Monet - Bend in the River Epte

1 History of the Haar Measure

In August of 1900 David Hilbert posed his famous list of 23 unsolved problems. The fifth problem asked for a topological description of Lie groups without any reference to smooth structure. In 1933 the Haar measure was first introduced by Alfred Haar in his paper *Der Massbegriff in der Theorie der kontinuierlichen Gruppen*. Haar established the existence of these measures on topological groups with a countable basis focusing on continuous groups like Lie groups. The theory was expanded significantly in the following decade. In 1963 John Von Neumann proved the uniqueness of the Haar measure up to scalar multiples. In 1940, Andre Weil proved the existence of the Haar measure for every locally compact topological group, removing the condition of a countable basis. The Haar measure arose as part of the machinery necessary to answer this problem.

2 Topological Preliminaries

The goal of these notes is to understand Haar measure. Haar measure is the natural analogue of Lebesgue measure on a locally compact group. The idea is that if a space has both a topology and a group structure, then we should be able to define a measure that respects translation by group elements. Before doing this, we recall the relevant topological language.

Definition 2.1 (Locally Compact). *A topological space X is said to be locally compact if for every point $x \in X$, there exists an open set U and a compact set K such that*

$$x \in U \subseteq K.$$

Example 2.1. *The space $\mathbb{R}^d$ is locally compact. Indeed, for each $x \in \mathbb{R}^d$, a sufficiently small open ball around x has compact closure.*

Definition 2.2 (Normal Topological Space). *A topological space X is normal if for any disjoint closed subsets $A, B \subset X$, there exist disjoint open sets U, V such that*

$$A \subseteq U, \quad B \subseteq V, \quad U \cap V = \emptyset.$$

Normality is useful because it allows us to construct continuous functions separating closed sets. In particular, it is the topological input behind Urysohn's lemma, which will later allow us to approximate characteristic functions by continuous compactly supported functions.

2.1 Nets

In metric spaces, many topological ideas can be understood using sequences. In general topological spaces, sequences are sometimes too weak. Nets are a generalization of sequences that work well in arbitrary topological spaces.

Definition 2.3 (Directed Set). *A directed set $(I, \leq)$ consists of a set I with a preorder $\leq$ such that for every $\alpha, \beta \in I$, there exists $\lambda \in I$ such that*

$$\alpha \leq \lambda \quad \text{and} \quad \beta \leq \lambda.$$

Example 2.2. *The natural numbers $\mathbb{N}$ with the usual ordering form a directed set. Thus every sequence is a special case of a net.*

Definition 2.4 (Net). *Given a topological space X , a net in X is a function*

$$I \rightarrow X, \quad \alpha \mapsto x_\alpha,$$

where $(I, \leq)$ is a directed set. We denote the net by $\{x_\alpha\}_{\alpha \in I}$.

Definition 2.5 (Convergent Net). *We say that a net $\{x_\alpha\}_{\alpha \in I}$ converges to $x \in X$ if for every neighborhood U of x , there exists $\alpha_0 \in I$ such that*

$$x_\alpha \in U$$

for all $\alpha \geq \alpha_0$.

3 Topological Groups

A topological group is a group whose algebraic structure is compatible with its topology. This compatibility is exactly what allows us to translate neighborhoods and eventually define a translation-invariant measure.

Definition 3.1 (Topological Group). *A group G with a binary operation*

$$G \times G \rightarrow G, \quad (x, y) \mapsto xy,$$

and inverse map

$$G \rightarrow G, \quad x \mapsto x^{-1},$$

is a topological group if its binary operation and inversion are continuous.

This semester we became intimately familiar with the topological group of real numbers $(\mathbb{R}^d, +)$. It will be useful throughout the presentation of topological groups to keep in mind the familiar example of real numbers. Observe that pairing a group with a discrete topology automatically makes any discrete group into a topological group. Lie groups, groups that are also finite dimensional real smooth manifolds, are defined with smooth binary operation and inversion automatically making them topological groups. These groups include the general linear groups of degree n in $GL_n(\mathbb{R})$, rotation matrices $SO(n, \mathbb{R})$, or the complex unitary group U_n .

Definition 3.2 (Symmetric Neighborhood). *A neighborhood V of 1 is called symmetric if*

$$V^{-1} = V.$$

Equivalently, $x \in V$ implies $x^{-1} \in V$.

Given $A \subset G$ and $x \in G$, define

$$Ax := \{yx : y \in A\}, \quad xA := \{xy : y \in A\}, \quad A^{-1} := \{y^{-1} : y \in A\}.$$

If $B \subset G$, define

$$AB = \{xy : x \in A, y \in B\}.$$

A key observation is that neighborhood problems in topological groups often reduce to neighborhoods of the identity. If U is an open neighborhood of 1, then xU is an open neighborhood of x , since $x = x1 \in xU$. We will dive into some interesting examples of topological groups below before fleshing out the theory necessary to define the Haar measure.

Example 3.1 (Circle Group). *The circle*

$$\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$$

is a compact topological group under multiplication. The topology and continuity of group operations are inherited from the topology of the complex plane.

Example 3.2 (Product Groups). *If $\{G_\alpha\}_{\alpha \in A}$ is a family of topological groups, then*

$$G = \prod_{\alpha \in A} G_\alpha$$

is a topological group with the product topology and coordinate-wise multiplication.

Example 3.3 (Cantor Group). *The Cantor group is defined as*

$$\prod_{n=1}^{\infty} \mathbb{Z}/2\mathbb{Z}$$

with the product topology. This group is homeomorphic to the standard Cantor set. Each cyclic group $\mathbb{Z}/2\mathbb{Z}$ is a discrete group which endowed with the discrete topology is automatically continuous. Then following from the previous example the product inherits the topological group structure with coordinate-wise binary operation modulo 2.

Example 3.4 (p -adic Integers). The p -adic integers $\mathbb{Z}_p$ which can be shown to be a topological group by probing of the p -adic measure. On the other hand the p -adic integers are also the inverse limit of cyclic groups which is a subgroup of the direct product of discrete cyclic groups $\mathbb{Z}/p^n\mathbb{Z}$. This means that we inherit the topological group structure of the direct product.

Proposition 3.1. Let G be a topological group.

- (a) The topology of G is invariant under translation and inversion. That is, if $U \subseteq G$ is open, then xU , Ux , and U^{-1} are open for all $x \in G$. Also, UA and AU are open for all $A \subseteq G$.
- (b) For every neighborhood U of 1, there is a symmetric neighborhood V of 1 such that

$$VV \subset U.$$

- (c) If H is a subgroup of G , then $\overline{H}$ is a subgroup of G .
- (d) Every open subgroup of G is closed.
- (e) If A and B are compact subsets of G , then AB is compact.

Proof. (a) For fixed $x \in G$, the maps

$$y \mapsto xy, \quad y \mapsto yx, \quad y \mapsto y^{-1}$$

are homeomorphisms of G . Therefore they send open sets to open sets. Hence xU , Ux , and U^{-1} are open.

Moreover,

$$AU = \bigcup_{a \in A} aU, \quad UA = \bigcup_{a \in A} Ua.$$

Since each translate is open, both AU and UA are open.

- (b) Let U be a neighborhood of 1. By continuity of multiplication at $(1, 1)$, there exist neighborhoods U_1 and U_2 of 1 such that

$$U_1U_2 \subset U.$$

Define

$$V = U_1 \cap U_2 \cap U_1^{-1} \cap U_2^{-1}.$$

Then V is symmetric, V is a neighborhood of 1, and

$$VV \subset U_1U_2 \subset U.$$

- (c) We use the subgroup test. Let $x, y \in \overline{H}$. Choose nets $\{x_\alpha\}$ and $\{y_\beta\}$ in H such that $x_\alpha \rightarrow x$ and $y_\beta \rightarrow y$. By continuity of multiplication and inversion,

$$x_\alpha y_\beta^{-1} \rightarrow xy^{-1}.$$

Since $x_\alpha y_\beta^{-1} \in H$, we have $xy^{-1} \in \overline{H}$. Hence $\overline{H}$ is a subgroup.

- (d) Let H be an open subgroup of G . Every left coset gH is open by part (a). Since

$$G \setminus H = \bigcup_{g \notin H} gH,$$

the complement of H is open. Thus H is closed.

- (e) The set AB is the image of $A \times B$ under the continuous multiplication map

$$G \times G \rightarrow G.$$

Since $A \times B$ is compact, its continuous image AB is compact.

□

4 Quotients of Topological Groups

The next natural object is the quotient group. Quotients are useful because they allow us to collapse a subgroup to the identity. Topologically, this is done by placing the quotient topology on the space of cosets.

Let H be a subgroup of a topological group G . The quotient G/H is the space of left cosets of H with the canonical quotient map

$$q : G \rightarrow G/H.$$

We give G/H the quotient topology:

$$U \subset G/H \text{ is open} \iff q^{-1}(U) \text{ is open in } G.$$

Proposition 4.1. *The canonical quotient map $q : G \rightarrow G/H$ takes open sets to open sets.*

Proof. Let $V \subset G$ be open. Then

$$q^{-1}(q(V)) = VH.$$

By the translation invariance of the topology, VH is open. Therefore $q(V)$ is open by definition of the quotient topology. $\square$

Proposition 4.2. *Let H be a subgroup of a topological group G .*

- (a) *If H is closed, then G/H is Hausdorff.*
- (b) *If G is locally compact, then G/H is locally compact.*
- (c) *If H is normal, then G/H is a topological group.*

Proof. (a) Let $q(x)$ and $q(y)$ be distinct points in G/H . Then $xH \neq yH$, so

$$y^{-1}x \notin H.$$

Since H is closed, there exists a neighborhood W of 1 such that

$$y^{-1}xW \cap H = \emptyset.$$

Choose a symmetric neighborhood V of 1 such that $VV \subset W$.

We claim that $q(xV)$ and $q(yV)$ are disjoint neighborhoods of $q(x)$ and $q(y)$. If not, then for some $v_1, v_2 \in V$,

$$xv_1H = yv_2H.$$

Thus

$$y^{-1}xv_1v_2^{-1} \in H.$$

But $v_1v_2^{-1} \in VV \subset W$, contradicting $y^{-1}xW \cap H = \emptyset$. Hence G/H is Hausdorff.

- (b) Let $q(x) \in G/H$. Since G is locally compact, choose an open set U and a compact set K such that

$$x \in U \subset K.$$

Then

$$q(x) \in q(U) \subset q(K).$$

The set $q(U)$ is open because q is an open map, and $q(K)$ is compact because q is continuous. Hence G/H is locally compact.

- (c) If H is normal, then G/H is a group. Multiplication and inversion are given by

$$q(x)q(y) = q(xy), \quad q(x)^{-1} = q(x^{-1}).$$

These operations are continuous because they are induced by the continuous multiplication and inversion maps on G together with the quotient topology. Therefore G/H is a topological group. $\square$

Proposition 4.3. *A topological group G is Hausdorff if and only if $\{1\}$ is closed.*

Proof. If G is Hausdorff, then every singleton is closed, so $\{1\}$ is closed.

Conversely, suppose $\{1\}$ is closed. Let $x, y \in G$ with $x \neq y$. Then $y^{-1}x \neq 1$. Since $\{1\}$ is closed, there exists a neighborhood W of 1 such that

$$y^{-1}xW \cap \{1\} = \emptyset.$$

Choose a symmetric neighborhood V of 1 such that $VV \subset W$. If $xV \cap yV \neq \emptyset$, then for some $v_1, v_2 \in V$,

$$xv_1 = yv_2.$$

Thus

$$y^{-1}x = v_2v_1^{-1} \in VV \subset W,$$

contradicting the choice of W . Hence xV and yV are disjoint neighborhoods of x and y . Therefore G is Hausdorff. $\square$

If a topological group is not Hausdorff, one often passes to the quotient

$$G/\overline{\{1\}}.$$

The subgroup $\overline{\{1\}}$ is closed and normal, and the quotient is Hausdorff. This signifies that it is essentially sufficient to study Hausdorff topological groups. When we say **locally compact group**, we mean a topological group whose topology is both locally compact and Hausdorff.

Definition 4.1 (σ -Compact). *A topological space is σ -compact if it is a countable union of compact sets.*

Proposition 4.4. *If G is a locally compact group, then G contains a subgroup H that is open, closed, and σ -compact.*

Proof. Since G is locally compact, choose an open neighborhood U of 1 such that $\overline{U}$ is compact. Replacing U by $U \cap U^{-1}$, we may assume that U is symmetric.

Define

$$H = \bigcup_{n=1}^{\infty} U^n,$$

where U^n denotes the product of n copies of U . Since U is symmetric and contains 1, the set H is a subgroup of G . Since each U^n is open, H is open. Every open subgroup is closed, so H is closed.

Finally,

$$H \subset \bigcup_{n=1}^{\infty} \overline{U}^n.$$

Each $\overline{U}^n$ is compact because it is the product of finitely many compact sets. Therefore H is σ -compact. $\square$

This proposition says that every locally compact group contains a large manageable piece: an open, closed, σ -compact subgroup. Since G is a disjoint union of cosets of H , the group can be viewed as a disjoint union of homeomorphic copies of a σ -compact space. This property is particularly useful as we see, because a Radon measure on a sigma-compact set is automatically both inner and outer regular.

5 Translations of Functions

We are eventually interested in developing a measure that will respect the translation invariance of open subsets of topological groups. It becomes useful to introduce translations at the level of functions and identify how translation acts on a function. For a function $f : G \rightarrow \mathbb{R}$, define

$$L_y f(x) = f(y^{-1}x), \quad R_y f(x) = f(xy).$$

Then

$$L_y L_x f = L_{yx} f.$$

Definition 5.1 (Left and Right Uniform Continuity). *A function f on a topological group G is left uniformly continuous if*

$$\|L_y f - f\|_{\sup} = \sup_{x \in G} |L_y f(x) - f(x)| \rightarrow 0$$

as $y \rightarrow 1$.

It is right uniformly continuous if

$$\|R_y f - f\|_{\sup} = \sup_{x \in G} |R_y f(x) - f(x)| \rightarrow 0$$

as $y \rightarrow 1$.

Proposition 5.1. *If $f \in C_c(G)$, then f is left and right uniformly continuous.*

Proof. We prove right uniform continuity. The proof for left uniform continuity is similar.

Let $K = \text{supp}(f)$ and fix $\varepsilon > 0$. For each $x \in K$, continuity of f gives a neighborhood U_x of 1 such that

$$|f(xu) - f(x)| < \frac{\varepsilon}{2}$$

whenever $u \in U_x$.

Choose a symmetric neighborhood V_x of 1 such that

$$V_x V_x \subset U_x.$$

Since the sets xV_x cover K , compactness gives $x_1, \dots, x_n \in K$ such that

$$K \subset \bigcup_{j=1}^n x_j V_{x_j}.$$

Let

$$V = \bigcap_{j=1}^n V_{x_j}.$$

If $y \in V$ and $x \in K$, choose j such that $x = x_j v$ with $v \in V_{x_j}$. Then $vy \in V_{x_j} V_{x_j} \subset U_{x_j}$, so

$$|f(xy) - f(x)| \leq |f(x_j vy) - f(x_j)| + |f(x_j) - f(x_j v)| < \varepsilon.$$

If $x \notin K$ and $xy \notin K$, then both $f(x)$ and $f(xy)$ are zero. If $x \notin K$ but $xy \in K$, apply the previous case to xy and use that $y^{-1} \in V$. Thus

$$\sup_{x \in G} |f(xy) - f(x)| < \varepsilon.$$

Hence f is right uniformly continuous. □

6 Radon Measures

We now move from topology to measure theory. The guiding example is the Lebesgue measure on $\mathbb{R}^d$. Lebesgue measure interacts well with the topology: open sets approximate measurable sets from outside, compact sets approximate many measurable sets from inside, and continuous compactly supported functions can be integrated. Radon measures axiomatize these properties on locally compact Hausdorff spaces. Throughout this section, X is assumed to be locally compact and Hausdorff.

6.1 Positive Linear Functionals on $C_c(X)$

Let $C_c(X)$ denote the set of real-valued continuous functions on X with compact support. This is a vector space over $\mathbb{R}$. We use the uniform norm

$$\|f\|_u = \sup_{x \in X} |f(x)|.$$

Definition 6.1 (Positive Linear Functional). *A linear functional I on $C_c(X)$ is a linear map*

$$I : C_c(X) \rightarrow \mathbb{R}.$$

It is called positive if

$$I(f) \geq 0$$

whenever $f \geq 0$.

Proposition 6.1. *If I is a positive linear functional on $C_c(X)$, then for each compact $K \subset X$, there is a constant C_K such that*

$$|I(f)| \leq C_K \|f\|_u$$

for all $f \in C_c(X)$ with $\text{supp}(f) \subset K$.

Proof. By Urysohn's lemma, choose $\phi \in C_c(X)$ such that

$$0 \leq \phi \leq 1 \quad \text{and} \quad \phi = 1 \text{ on } K.$$

If $\text{supp}(f) \subset K$, then

$$|f| \leq \|f\|_u \phi.$$

Hence

$$\|f\|_u \phi - f \geq 0, \quad \|f\|_u \phi + f \geq 0.$$

Applying positivity of I gives

$$\|f\|_u I(\phi) - I(f) \geq 0, \quad \|f\|_u I(\phi) + I(f) \geq 0.$$

Therefore

$$|I(f)| \leq I(\phi) \|f\|_u.$$

Taking $C_K = I(\phi)$ gives the result. □

Definition 6.2 (Radon Measure). *A Borel measure μ on a locally compact Hausdorff space X is called a Radon measure if:*

1. **Outer regularity:**

$$\mu(E) = \inf\{\mu(U) : E \subset U, U \text{ open}\}.$$

2. **Inner regularity:**

$$\mu(E) = \sup\{\mu(K) : K \subset E, K \text{ compact}\}$$

for Borel sets E .

3. **Local finiteness:**

$$\mu(K) < \infty$$

for every compact set $K \subset X$.

A familiar example of a Radon Measure is the Lebesgue measure which we have already shown to satisfy properties 1-3 on the locally compact Hausdorff space $\mathbb{R}$. If μ is a Radon measure, then every function $f \in C_c(X)$ is integrable and

$$f \mapsto \int_X f d\mu$$

is a positive linear functional on $C_c(X)$.

Theorem 6.1 (Riesz Representation Theorem). *Let X be a locally compact Hausdorff space. If I is a positive linear functional on $C_c(X)$, then there exists a unique Radon measure μ on X such that*

$$I(f) = \int_X f d\mu$$

for every $f \in C_c(X)$.

The proof of this theorem is beyond the scope of this paper but it does outline the path we need to travel in topological groups in order to define a measure. As we will see in the next section, the Haar measure will be the result of a Radon measure that plays well with the translation invariance of the topological group. In order to show that this measure exists on any locally compact topological group we will start by constructing a special positive linear functional and then invoking the Riesz Representation Theorem.

7 Haar Measure

We now specialize to locally compact groups. The point of Haar measure is to choose a Radon measure that is compatible with the group operation. While we haven't yet guaranteed the existence we will begin under the assumption that we can define a unique Radon measure on a locally compact group to outline some properties of the Haar measure and then return in the following section to prove that it always exists on a locally compact topological group.

Definition 7.1 (Left and Right Haar Measure). *Let G be a locally compact group. A nonzero Radon measure μ on G is called a left Haar measure if*

$$\mu(xE) = \mu(E)$$

for every Borel set $E \subset G$ and every $x \in G$.

It is called a right Haar measure if

$$\mu(Ex) = \mu(E)$$

for every Borel set $E \subset G$ and every $x \in G$.

Definition 7.2 (Invariant Functional). *A positive linear functional I on $C_c(G)$ is left invariant if*

$$I(L_x f) = I(f)$$

for every $x \in G$ and every $f \in C_c(G)$.

The connection between measures and functionals is that a left Haar measure gives a left-invariant positive linear functional:

$$I(f) = \int_G f d\mu.$$

Conversely, by the Riesz representation theorem, a left-invariant positive linear functional gives a left Haar measure.

Proposition 7.1. *Let G be a locally compact group.*

(a) *A Radon measure μ on G is a left Haar measure if and only if the measure $\tilde{\mu}$ defined by*

$$\tilde{\mu}(E) = \mu(E^{-1})$$

is a right Haar measure.

(b) *A nonzero Radon measure μ on G is a left Haar measure if and only if*

$$\int_G L_y f d\mu = \int_G f d\mu$$

for every $f \in C_c(G)$ and every $y \in G$.

(c) *If μ is a left Haar measure on G , then $\mu(U) > 0$ for every nonempty open set $U \subset G$ and*

$$\int_G f d\mu > 0$$

for every nonzero $f \in C_c(G)$ with $f \geq 0$.

(d) *If μ is a left Haar measure on G , then*

$$\mu(G) < \infty$$

if and only if G is compact.

Proof. (a) The inversion map $x \mapsto x^{-1}$ is a homeomorphism of G . Therefore it takes Borel sets to Borel sets, open sets to open sets, and compact sets to compact sets. Hence $\tilde{\mu}$ is again a Radon measure.

For a Borel set E and $g \in G$,

$$\tilde{\mu}(Eg) = \mu((Eg)^{-1}) = \mu(g^{-1}E^{-1}).$$

Thus $\tilde{\mu}(Eg) = \tilde{\mu}(E)$ for all E and g if and only if

$$\mu(g^{-1}E^{-1}) = \mu(E^{-1})$$

for all E and g , which is precisely left invariance of μ .

(b) Suppose first that μ is left Haar. Let h be a nonnegative simple function,

$$h = \sum_{i=1}^n \alpha_i \mathbb{1}_{E_i}.$$

Then

$$L_g h(x) = h(g^{-1}x) = \sum_{i=1}^n \alpha_i \mathbb{1}_{gE_i}(x).$$

Hence

$$\int_G L_g h \, d\mu = \sum_{i=1}^n \alpha_i \mu(gE_i) = \sum_{i=1}^n \alpha_i \mu(E_i) = \int_G h \, d\mu.$$

By approximation by simple functions, this holds for all nonnegative measurable functions, and hence for every $f \in C_c(G)$ by decomposing $f = f^+ - f^-$.

Conversely, suppose

$$\int_G L_y f \, d\mu = \int_G f \, d\mu$$

for every $f \in C_c(G)$ and every $y \in G$. Let $U \subset G$ be open. By regularity and Urysohn's lemma, the value $\mu(U)$ can be recovered from compactly supported continuous functions:

$$\mu(U) = \sup \left\{ \int_G f \, d\mu : f \in C_c(G), 0 \leq f \leq \mathbb{1}_U \right\}.$$

Since $0 \leq f \leq \mathbb{1}_U$ if and only if $0 \leq L_{g^{-1}} f \leq \mathbb{1}_{gU}$, the assumed integral invariance gives

$$\mu(gU) = \mu(U)$$

for every open set U and every $g \in G$. By outer regularity, this extends to all Borel sets. Therefore μ is left Haar.

(c) Since μ is nonzero, there exists a Borel set E with $\mu(E) > 0$. By inner regularity, there exists a compact set $K \subset E$ with $\mu(K) > 0$.

Let U be a nonempty open subset of G . Since the translates of U cover G , they cover K . Compactness of K gives

$$K \subset \bigcup_{j=1}^n x_j U.$$

If $\mu(U) = 0$, then by left invariance,

$$\mu(K) \leq \sum_{j=1}^n \mu(x_j U) = 0,$$

contradiction. Hence $\mu(U) > 0$.

If $f \in C_c(G)$, $f \geq 0$, and $f \neq 0$, then there exists $c > 0$ such that

$$U = \{x \in G : f(x) > c\}$$

is nonempty and open. Therefore

$$\int_G f d\mu \geq c\mu(U) > 0.$$

(d) If G is compact, then $\mu(G) < \infty$ by local finiteness.

Conversely, suppose $\mu(G) < \infty$. Let V be a neighborhood of 1 with compact closure. Choose a symmetric neighborhood U of 1 such that

$$UU \subset V.$$

If finitely many left translates of V did not cover G , we could choose a sequence $\{x_n\}$ such that

$$x_m \notin \bigcup_{i=1}^{m-1} x_i V.$$

Then the sets $x_n U$ are pairwise disjoint. Indeed, if $x_n U \cap x_m U \neq \emptyset$ with $m > n$, then for some $u_0, u_1 \in U$,

$$x_n u_1 = x_m u_0.$$

Hence

$$x_m = x_n u_1 u_0^{-1} \in x_n U U \subset x_n V,$$

contradiction.

Since $\mu(U) > 0$ by part (c), left invariance gives

$$\mu(G) \geq \sum_{n=1}^{\infty} \mu(x_n U) = \sum_{n=1}^{\infty} \mu(U) = \infty,$$

contradiction. Thus finitely many translates of V cover G . Since $\bar{V}$ is compact, finitely many translates of $\bar{V}$ cover G , so G is compact. □

8 Existence of Haar Measure

We now sketch the construction of Haar measure. The basic idea is to measure a set by counting how many small translates are needed to cover it. This resembles the construction of Lebesgue outer measure from coverings by intervals, except that the intervals are replaced by translates of neighborhoods of the identity.

For an open neighborhood V of 1 and a set $E \subset G$, define

$$(E : V) = \inf \left\{ \#(A) : E \subset \bigcup_{x \in A} xV \right\},$$

where $\#(A) = |A|$ if A is finite, and $\#(A) = \infty$ otherwise. The number $(E : V)$ is the minimum number of left translates of V needed to cover E .

It is technically more convenient to work with continuous compactly supported functions rather than sets.

Let

$$C_c^+(G) = \{f \in C_c(G) : f \geq 0, f \neq 0\}.$$

Lemma 8.1. *If $f, \phi \in C_c^+(G)$, then there exist $x_1, \dots, x_n \in G$ such that*

$$f \leq \frac{2\|f\|_u}{\|\phi\|_u} \sum_{i=1}^n L_{x_i} \phi.$$

Proof. Since $\phi \neq 0$, the set

$$W = \left\{ x \in G : \phi(x) > \frac{1}{2} \|\phi\|_u \right\}$$

is nonempty and open. The compact set $\text{supp}(f)$ can be covered by finitely many left translates of W :

$$\text{supp}(f) \subset \bigcup_{i=1}^n x_i W.$$

On each $x_i W$, the function $L_{x_i} \phi$ is greater than $\frac{1}{2} \|\phi\|_u$. Therefore the displayed inequality follows. $\square$

Definition 8.1 (Haar Covering Number). *For $f, \phi \in C_c^+(G)$, define*

$$(f : \phi) = \inf \left\{ \sum_{j=1}^n c_j : f \leq \sum_{j=1}^n c_j L_{x_j} \phi, \ c_j > 0, \ x_j \in G \right\}.$$

Lemma 8.2. *Suppose $f, g, \phi \in C_c^+(G)$.*

- (a) $(L_x f : \phi) = (f : \phi)$ for every $x \in G$.
- (b) $(c f : \phi) = c(f : \phi)$ for every $c > 0$.
- (c) $(f + g : \phi) \leq (f : \phi) + (g : \phi)$.
- (d) $(f : \phi) \leq (f : g)(g : \phi)$.

Proof. (a) If

$$f \leq \sum_{j=1}^n c_j L_{x_j} \phi,$$

then applying L_x gives

$$L_x f \leq \sum_{j=1}^n c_j L_{x x_j} \phi.$$

Thus $(L_x f : \phi) \leq (f : \phi)$. Applying the same argument to $L_x f$ with x^{-1} gives the reverse inequality.

- (b) This follows by multiplying admissible inequalities by c and by dividing admissible inequalities for $c f$ by c .
- (c) Add an admissible inequality for f to an admissible inequality for g , then take infima.
- (d) If

$$f \leq \sum_j c_j L_{x_j} g$$

and

$$g \leq \sum_k d_k L_{y_k} \phi,$$

then

$$f \leq \sum_{j,k} c_j d_k L_{x_j y_k} \phi.$$

Taking infima gives the claim. $\square$

Choose a fixed function $f_0 \in C_c^+(G)$ and define

$$I_\phi(f) = \frac{(f : \phi)}{(f_0 : \phi)}.$$

The functionals I_ϕ are normalized and left invariant, but they are only subadditive. As the support of ϕ shrinks toward the identity, compactness arguments allow one to extract a limit that becomes additive.

Lemma 8.3. *If $f_1, f_2 \in C_c^+(G)$ and $\varepsilon > 0$, there is a neighborhood V of e such that $I_\phi(f_1) + I_\phi(f_2) \leq I_\phi(f_1 + f_2) + \varepsilon$ whenever $\text{supp}(\phi) \subset V$.*

Proof. Fix $g \in C_c^+(G)$ such that $g = 1$ on $\text{supp}(f_1 + f_2)$, and let $\delta > 0$. Set $h = f_1 + f_2 + \delta g$ and $h_1 = f_1/h$ and $h_2 = f_2/h$ where both functions are zero outside $\text{supp}(f_1)$ and $\text{supp}(f_2)$ respectively. Then $h_i \in C_c^+(G)$, then there is a neighborhood V of the identity e on which f is left and right uniformly continuous. That is $|h_i(x) - h_i(y)| < \delta$ for $i = 1, 2$ and $y^{-1}x \in V$. If $\phi \in C_c^+(G)$, $\text{supp}(\phi) \subset V$ and $h \leq \sum_{j=1}^n c_j L_{x_j} \phi$ then $|h_i(x) - h_i(y)| < \delta$ whenever $x_j^{-1}x \in \text{supp}(\phi)$. Then,

$$f_i(x) = h(x)h_i(x) \leq \sum_j c_j \phi(x_j^{-1}x)h_i(x) \leq \sum_j c_j \phi(x_j^{-1}x)[h_i(x_j) + \delta]$$

Then, $(f_i : \phi) \leq \sum c_j [h_i(x_j) + \delta]$. Since $h_1 + h_2 \leq 1$,

$$(f_1 : \phi) + (f_2 : \phi) \leq \sum_j c_j [1 + 2\delta]$$

By minimizing c_j we can get arbitrarily close to $(h : \phi)$. Then applying (2) and (3) of the previous set of lemmas,

$$I_\phi(f_1) + I_\phi(f_2) \leq (1 + 2\delta)I_\phi(h) \leq (1 + 2\delta)[I_\phi(f_1 + f_2) + \delta I_\phi(g)]$$

Be (4) of the above lemmas

$$(f_0 : f)^{-1} \leq I_\phi(f) \leq (f : f_0)$$

Then we may choose δ sufficiently small such that

$$2\delta(f_1 + f_2 : f_0) + \delta(1 + 2\delta)(g : f_0) < \varepsilon$$

and we are done. □

With this result established we are ready to show the main proof of this paper.

Theorem 8.1. *Every locally compact group G possess a left Haar Measure.*

Proof. For each $f \in C_c^+(G)$ let $X_f := [(f_0 : f)^{-1} : (f : f_0)]$ and define

$$X = \prod_{f \in C_c^+(G)} X_f$$

Since each X_f is compact, the product X is compact with respect to the product topology. Since $(f_0 : f)^{-1} \leq I_\phi(f) \leq (f : f_0)$ every I_ϕ is an element of X . For each compact neighborhood V of the identity e . Let $K(V)$ be the closure in X of $\{I_\alpha : \text{supp}(\phi) \subset V\}$. It follows that

$$K(\cap_i^n V_i) \subset \cap_i^n K(V_i)$$

Since each V_i is compact, their intersection is compact and must be non-empty. Every neighborhood of I in X intersects $\{I_\alpha : \text{supp}(\phi) \subset V\}$ for all of V . This means that for any compact neighborhood V of e and any $f_1, \dots, f_n \in C_c^+(G)$ and for all $\varepsilon > 0$ there exists a $\phi \in C_c^+(G)$ with $\text{supp}(\phi) \subset V$ such that $|I(f_i) - I_\phi(f_i)| < \varepsilon$. Combined with the two previous lemmas this implies that I is left-invariant and that

$$I(af + bg) = aI(f) + bI(g)$$

for all $f, g \in C_c^+(G)$ and $a, b > 0$. By setting $I(f) = I(f^+) - I(f^-)$, the function becomes a left-invariant positive linear functional on $C_c(G)$. Then the proof is completed by invocation of the Riesz Representation Theorem. □

9 Uniqueness of Haar Measure

The existence theorem says that every locally compact group has a left Haar measure. The next theorem says that this measure is essentially unique: the only freedom is multiplication by a positive constant.

Theorem 9.1 (Uniqueness of Haar Measure). *Let μ and ν be two left Haar measures on a locally compact group G . Then there exists a constant $c > 0$ such that*

$$\nu = c\mu.$$

Proof. The statement that $\mu = c\nu$ is equivalent to the assertion that the ration

$$r_f = \frac{\int f d\mu}{\int f d\nu}$$

is independent of $f \in C_c^+(G)$. Suppose that $f, g \in C_c^+(G)$ then we will show that $r_f = r_g$. Fix a symmetric compact neighborhood V_0 of e and set

$$A = [\text{supp}(f)]V_0 \cup V_0[\text{supp}(f)] \quad B = [\text{supp}(g)]V_0 \cup V_0[\text{supp}(g)]$$

Since V_0 and the support of each function are compact then A and B are compact. For $y \in V_0$ the functions

$$x \mapsto f(xy) - f(yx) \quad \text{and} \quad x \mapsto g(xy) - g(yx)$$

are supported on A and B respectively. For all $\varepsilon > 0$, there exists a symmetric compact neighborhood $V \subset V_0$ of e such that

$$\sup_x |f(xy) - f(yx)| < \varepsilon \quad \text{and} \quad \sup_x |g(xy) - g(yx)| < \varepsilon$$

for $y \in V$. Pick $h \in C_c^+(G)$ with $\text{supp}(h) \subset V$ and $h(x) = h(x^{-1})$. Then,

$$\int h d\nu \int f d\mu = \int \int h(y)f(x) d\mu(x) d\nu(y) \tag{1}$$

$$= \int \int h(y)f(yx) d\mu(x) d\nu(y) \tag{2}$$

Since we chose h such that $h(x) = h(x^{-1})$. Then,

$$\int h d\mu \int f d\nu = \int \int h(y)f(x) d\mu(x) d\nu(y) \tag{3}$$

$$= \int \int h(y^{-1}x)f(y) d\mu(x) d\nu(y) \tag{4}$$

$$= \int \int h(yx^{-1})f(y) d\mu(x) d\nu(y) \tag{5}$$

$$= \int \int h(y)f(yx) d\mu(x) d\nu(y) \tag{6}$$

Then,

$$\left| \int h d\nu \int f d\mu - \int h d\mu \int f d\nu \right| = \left| \int \int h(y)(f(xy) - f(yx)) d\mu(x) d\nu(y) \right| \tag{7}$$

$$\leq \varepsilon \mu(A) \int h d\nu \tag{8}$$

Similarly

$$\left| \int h d\nu \int g d\mu - \int h d\mu \int g d\nu \right| \leq \varepsilon \mu(B) \int h d\nu$$

Then dividing on both sides we obtain

$$\left| \frac{\int f d\mu}{\int f d\nu} - \frac{\int g d\mu}{\int g d\nu} \right| \leq \varepsilon \left(\frac{\mu(A)}{\int f d\nu} + \frac{\mu(B)}{\int g d\nu} \right)$$

Taking epsilon to zero then we have shown that $r_f = r_g$ and therefore the measures differ only by a constant. $\square$

Remark 9.1. If G is compact, it is common to normalize Haar measure by requiring

$$\mu(G) = 1.$$

With this normalization, Haar measure is unique rather than merely unique up to a scalar.

10 Simple Examples of Haar Measure

Example 10.1 (Lebesgue Measure as Haar Measure). The primary example is the additive group $(\mathbb{R}^d, +)$.

A left translate of a set $E \subset \mathbb{R}^d$ by a point $x \in \mathbb{R}^d$ is

$$x + E = \{x + y : y \in E\}.$$

Lebesgue measure satisfies

$$m(x + E) = m(E)$$

for every Lebesgue measurable set E and every $x \in \mathbb{R}^d$. Therefore Lebesgue measure is a Haar measure on the locally compact group $(\mathbb{R}^d, +)$.

Conversely, the uniqueness theorem says that every Haar measure on $(\mathbb{R}^d, +)$ is a positive scalar multiple of Lebesgue measure.

Example 10.2 (Counting Measure on a Discrete Group). Let G be a discrete group. Counting measure

$$\mu(E) = \#E$$

is a Haar measure. Indeed, left multiplication by a fixed element is a bijection, so it preserves the number of elements:

$$\#(xE) = \#E.$$

Example 10.3 (Finite Groups). If G is a finite group, then counting measure is Haar measure. The normalized Haar measure is

$$\mu(E) = \frac{\#E}{\#G}.$$

This is the unique Haar measure satisfying $\mu(G) = 1$.

Example 10.4 (The Circle Group). On the circle group

$$\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\},$$

Haar measure is normalized arc length measure. Rotation of the circle does not change arc length, so the measure is translation-invariant in the group sense.

Example 10.5 (Compact Groups). If G is compact, Haar measure can be normalized so that

$$\mu(G) = 1.$$

Thus compact groups naturally carry a probability measure that is invariant under left translation.

11 Further Topics

11.1 Unimodular Groups

A locally compact group is called unimodular if the left and right Haar measures agree up to the same normalization; equivalently, a left Haar measure is also right-invariant. Compact groups, discrete groups, abelian locally compact groups, and semisimple Lie groups are standard examples of unimodular groups.

Unimodularity matters because it is exactly the condition under which integration over the group behaves symmetrically with respect to left and right translations. This simplifies harmonic analysis, representation theory, ergodic theory, and the theory of lattices. For instance, if G is unimodular and $\Gamma \leq G$ is a lattice, then the quotient G/Γ carries a natural G -invariant measure. In representation theory, unimodularity is part of what makes convolution algebras and unitary representations behave cleanly: the formulae for adjoints, convolution, and integration over orbits are much nicer when there is no modular-function correction factor.

11.2 Hilbert's Fifth Problem

Haar measure also plays an important role in the background of Hilbert's fifth problem, which asked whether every locally Euclidean topological group must be a Lie group. More broadly, the solution developed a structure theory for locally compact groups, and Haar measure was one of the key analytic tools that made such groups accessible. Since Haar measure exists on every locally compact group, it gives a way to average functions, construct invariant objects, and bring measure-theoretic and functional-analytic methods into a setting that initially has only topology and algebra.

In the Gleason–Montgomery–Zippin solution of Hilbert's fifth problem the Haar measure supports the analytic machinery used to study small subgroups, quotient groups, and local group actions. The existence of invariant measure allows one to pass from local compactness to regularity properties strong enough to prove that a locally compact group with no small subgroups has the structure of a Lie group. The Haar measure helped bridge the gap between abstract topological groups and smooth Lie theory, making it one of the foundational tools in the modern structure theory of locally compact groups.

References

Gerald B. Folland. *Real Analysis Modern Techniques and their Applications*. A Wiley Interscience Publication, 2 edition, 1999.

Terrence Tao. *Hilbert's fifth problem and related topics*.